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Spike-driven Discrete Aggregation for Event-based Object Detection

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Abstract

With their high dynamic range and temporal resolution, event cameras are well-suited for object detection, especially under motion blur and extreme illumination. Recent state-of-the-art works for event-based object detection primarily focus on the high-level design of backbones. However, developing effective event representations is equally crucial, as it bridges asynchronous event streams with the dense tensors required by detection networks. Most existing aggregation strategies for event representation continuously accumulate all events within sampled intervals without selective filtering, inevitably introducing uninformative events that degrade detection accuracy. To address this limitation, we introduce a novel perspective, termed Discrete Aggregation, which adaptively and discretely selects informative events for differentiable aggregation. We realize this through the Spiking Discrete Aggregation (SDA) module, which is inspired by the threshold-based spike firing mechanism in Spiking Neural Networks (SNNs) and implemented using gated recurrent spiking neurons. Additionally, we introduce the Multi-Timescale Fusion (MTF) method which leverages coarse-grained temporal features from continuous event streams to further enhance the representation capability of SDA. Experimental results on neuromorphic datasets demonstrate that our method achieves state-of-the-art performance among all fully spiking architectures while using fewer parameters, reaching 43.4% $mAP_{50:95}$ on Gen1 (+ 4.5% over prior art). Moreover, our method exhibits superior robustness under noisy conditions and shows strong compatibility with non-spiking models.

1. Introduction

Unlike frame-based sensors, bio-inspired event cameras asynchronously capture pixel-wise brightness changes, offering high dynamic range ($>$ 120 dB) and microsecond-level temporal resolution ($>$ 10k fps) [10]. These characteristics are well-suited for object detection, scene attribute estimation [4, 5, 42], scene reconstruction [6, 21, 22], and video restoration and enhancement [7, 38, 40], especially in high-speed or extreme lighting scenarios.

Event streams contain rich spatiotemporal cues and motion information. To fully exploit these properties, recent state-of-the-art works on object detection [11, 13, 27–29, 39] have mainly focused on the high-level design of backbone, adopting only simple event representations such as Event Count [25]. However, effective event representation remains equally important, as the asynchronous and sparse nature of event data makes it incompatible with conventional deep learning models. Typically, event represen-

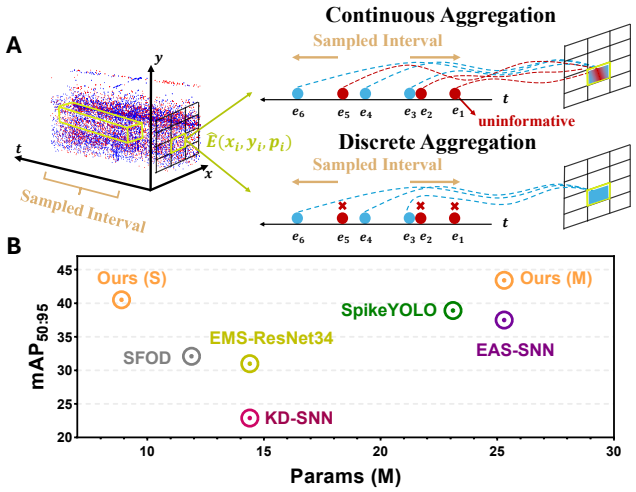


Figure 1. **A.** Illustration of continuous and discrete aggregation. Circles denote individual events and red circles indicate uninformative events. Continuous aggregation integrates all events within the sampled interval, whereas discrete aggregation selectively retains informative ones. **B.** Comparison of performance and model size on Gen1 between our method and state-of-the-art fully spiking detectors. M and S in the parentheses (·) indicate medium and small scales, respectively.

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tations are obtained by first sampling events into intervals and then aggregating events within each interval into dense tensors. We notice that existing event aggregation methods [3, 8, 18, 20, 24, 25, 36] integrate all events occurring within the sampled interval without any selective filtering, as shown in Fig. 1A. For example, ASTMNet [18] employs TCN to process per-pixel event sequences in an integrated manner; EAS-SNN [36] accumulates the membrane potentials computed for each event by spiking neurons, even when an event does not activate the neuron to fire a spike. We refer to this continuous accumulation of events over time as continuous aggregation. This process inevitably embeds redundant or irrelevant events (*e.g.*, sensor-induced artifacts and motion-related blur [12, 37]). To address this limitation, we propose a novel perspective: **discrete aggregation**, which adaptively and discretely retains informative events, unlike continuous aggregation (see Fig. 1A). Moreover, the aggregation mechanism can be optimized using high-level semantics from downstream networks.

To implement this idea, we propose the Spiking Discrete Aggregation (SDA) module, inspired by the spike-firing mechanism of Spiking Neural Networks (SNNs). In SNNs, a spike is generated only when the membrane potential reaches a certain threshold, which naturally aligns with discrete aggregation: irrelevant events fail to trigger spikes due to insufficient membrane potential accumulation, whereas informative events can activate spiking neurons. SDA employs pixel-wise gated recurrent spiking neurons to aggregate events that cause their corresponding neurons to fire spikes, and it can be optimized by downstream fully spiking detectors.

Although SDA excels at fine-grained event selection, event-based object detection also requires multi-timescale feature modeling, since the temporal density of events generally correlates with object motion speed (*e.g.* faster objects generate denser events, whereas slower ones result in sparser event distributions). Therefore, we introduce the Multi-Timescale Fusion (MTF) method, which incorporates coarse-grained temporal information from extended time intervals to enhance the representation learned by SDA. As shown in Fig. 1B, our detector equipped with SDA-MTF achieves superior performance while using fewer parameters compared to other fully spiking detectors.

The main contributions are summarized as follows:

- We propose a novel spike-driven discrete aggregation module for event-based object detection, implemented with gated recurrent spiking neurons. This module tightly couples the spike-firing mechanism of spiking neurons with the adaptive aggregation of informative events.
- We introduce the Multi-Timescale Fusion (MTF) method, which captures coarse-grained temporal features to enhance the representation capability of SDA.
- Our proposed method outperforms state-of-the-art fully spiking detectors on the Gen1, 1Mpx and N-Caltech101 datasets, while achieving 3.79× higher energy efficiency compared to ANNs. In addition, its effectiveness is further validated on various non-spiking architectures and under noisy conditions.

2. Related Work

2.1. Event Representation

Existing event representations can be broadly categorized into handcrafted and learning-based. Handcrafted methods, such as Event Count [25], Voxel Cube [8] and Time Surface [33], apply predefined rules [1, 20, 26, 31] to continuously convert event streams into dense formats, suffering from noise sensitivity and lack of adaptability. A notable exception is HOTS [17], which operates in a discrete manner by selectively extracting spatiotemporal features via a time-surface decay mechanism. However, its non-differentiable nature prevents end-to-end optimization. On the other hand, learning-based methods leverage neural networks to adaptively generate representations through data-driven optimization. This category encompasses (1) sparse representations, such as GNN-based [19, 32] and SNN-based [36, 41], whose sparse computational properties enable lower energy consumption but lead to poor performance; (2) dense representations [3, 18] which recently use temporal architectures to effectively extract spatiotemporal features, yet suffer from accumulation of uninformative events due to continuous aggregation. In this work, we propose a novel differentiable discrete aggregation approach that addresses these limitations by leveraging the spike-driven nature of SNNs.

2.2. Event-based Object Detection with SNNs

SNNs have been increasingly adopted in event-based object detection. EMS-ResNet [34] designs a full-spike residual block for detection. SpikeYOLO [23] explores a suitable detection architecture and proposes an SNN with integer-valued training and spike-driven inference. These works focus on architectural design but do not explicitly address effective low-level processing of event data. EAS-SNN [36] adaptively samples events between spike firing times for representation to mitigate the temporal mismatch between the raw event stream and SNNs. CREST [24] introduces a conjointly trained spike-driven framework with multi-scale spatiotemporal event representation for efficient event-based object detection. However, they still include uninformative events during continuous aggregation.

3. Preliminary

Event Representation for Detection. A sequence of events captured by event cameras is denoted as $E = \{e_i | e_i = (x_i, y_i, t_i, p_i)\}_{i \in N}$, where $x_i \in [1, W]$ and

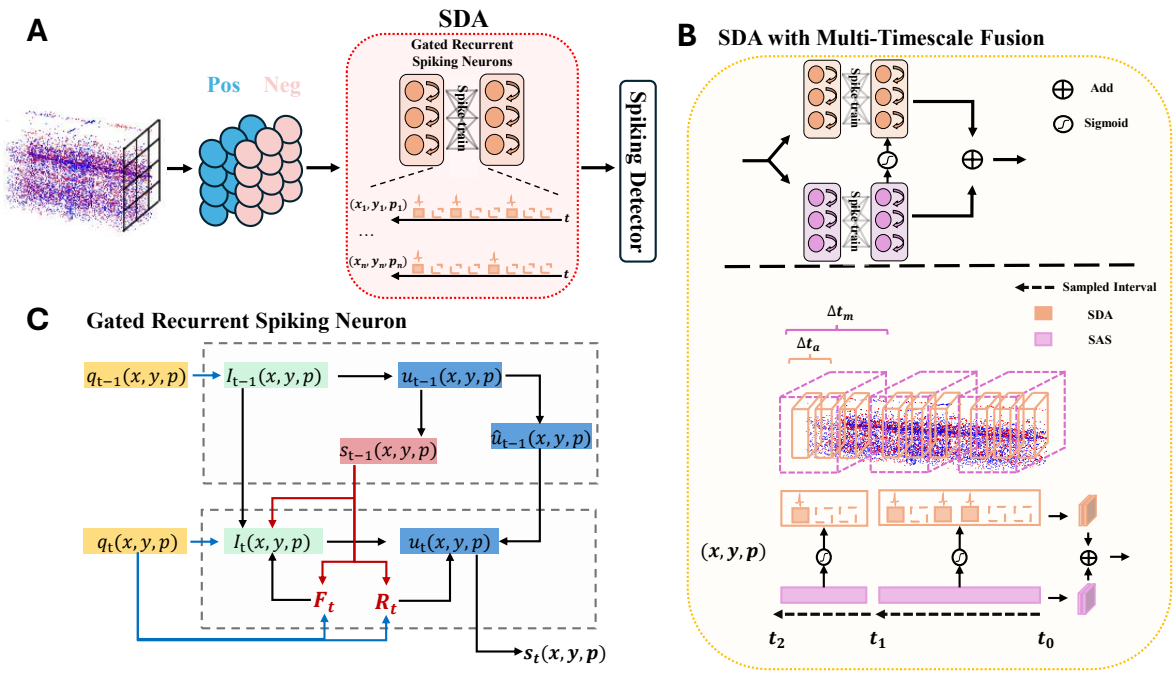


Figure 2. Overview of our approach. **A. Graphical illustration of the pipeline.** Event streams are first divided into positive and negative polarities and processed by early aggregation. The inputs are then fed into the Spiking Discrete Aggregation (SDA) module, which employs gated recurrent spiking neurons. SDA produces the representation that can be seamlessly integrated into a fully spiking detector. **B. SDA with Multi-Timescale Fusion.** The top part shows the architecture, where an additional timescale branch is incorporated based on the vanilla SDA. The bottom part presents a more detailed and intuitive view of the computation. **C. Dynamics of Gated Recurrent Spiking Neuron.** The red solid lines indicate the recurrent connections while the blue and black solid lines depict the input pathways and internal dynamics, respectively. R and F indicate the reset and forget gate.

$y_i \in [1, H]$ are spatial coordinates, $t_i \in [0, +\infty)$ is the timestamp, and $p_i \in \{0, 1\}$ indicates the polarity of brightness change. In the Gen1 and 1Mpx datasets, an annotation at a given time t is represented by $B = \{b_i | b_i = (x_i, y_i, w_i, h_i, l_i, t)\}_{i \in N}$, where (x_i, y_i, w_i, h_i) defines the bounding box and l_i denotes the object category.

Due to the hardware limitations in processing microsecond-resolution data and the lack of necessity to handle every event for object detection [29], most methods first sample the event stream using an operation $\mathcal{S}$, then aggregate the sampled events into a dense tensor H via an aggregator $\mathcal{A}$, and finally feed H into a detector $\mathcal{D}$:

$$\mathcal{D}(E) \approx \mathcal{D}(H) = \mathcal{D}(\mathcal{A}(\mathcal{S}(E))) \tag{1}$$

Taking fixed time-window sampling as an example, let the window size be Δt. The resulting dense tensor H_j aggregated by Event Count, a typical aggregation method, can be formulated as:

$$H_j = \mathcal{A}(\mathcal{S}_j(E)) = \mathcal{A}(\{e_i | t_i \in [(j-1)\Delta t, j\Delta t)\}) \tag{2}$$

$$\mathcal{A}(\hat{E}_j) = h(x, y, p) = \sum_{e_i \in \hat{E}_j} \delta(x - x_i, y - y_i, p - p_i) \tag{3}$$

where j is the index of the time interval in the global window, $\hat{E}_j$ denotes the set of sampled events, and $\delta(\cdot)$ is the Dirac delta function.

Leaky Integrate-and-Fire Neuron. Leaky Integrate-and-Fire (LIF) neuron is popular to construct SNNs for its biological plausibility and low computational cost. In practice, the iterative LIF equations are written as:

$$\boldsymbol{u}_i^l[t] = \tau \hat{\boldsymbol{u}}_i^l[t-1] + \boldsymbol{I}_i^l[t] \tag{4}$$

$$\boldsymbol{s}_i^l[t] = \Theta(\boldsymbol{u}_i^l[t] - V_{thresh}) \tag{5}$$

$$\hat{\boldsymbol{u}}_i^l[t] = u_{reset} \boldsymbol{s}_i^l[t] + \boldsymbol{u}_i^l[t](1 - \boldsymbol{s}_i^l[t]) \tag{6}$$

where t is the time step, τ denotes a constant decay factor, $\boldsymbol{u}_i^l[t]$ and $\boldsymbol{I}_i^l[t]$ denote membrane potential and input current of the i-th neuron in the l-th layer. Usually, $\boldsymbol{I}[t]$ is transformed by a parameterized mapping, such as convolution.

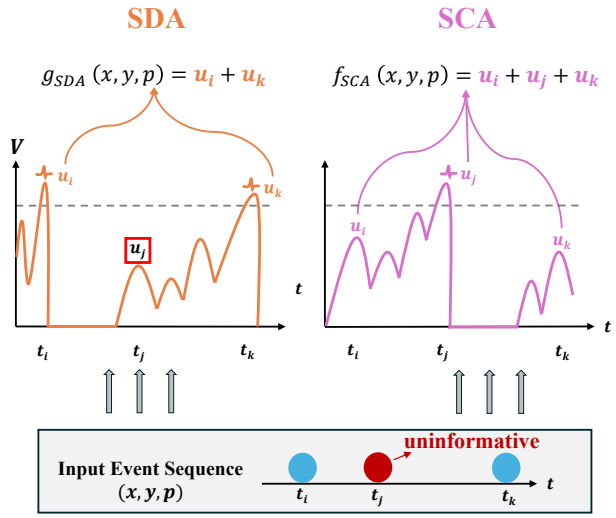


Figure 3. Spiking Discrete Aggregation. Different from SCA, SDA aggregates only the events that activate spiking neurons, effectively filtering out uninformative ones.

Θ is the Heaviside function. If $\boldsymbol{u}[t]$ exceeds membrane potential threshold V_{thresh}, the neuron triggers a spike $\boldsymbol{s}[t]$ and resets its potential to $\hat{\boldsymbol{u}}[t]$.

4. Method

Our motivation is to design a differentiable aggregation module that can (1) discretely and adaptively aggregate informative events after sampling, (2) fully exploit spatiotemporal correlations in event data, and (3) effectively capture temporal features across multiple timescales to further enhance the representation capability. Sections 4.1 to 4.3 detail these points. Fig. 2A shows an overview of the pipeline.

4.1. Spiking Discrete Aggregation

In this section, we introduce Spiking Discrete Aggregation (SDA), a differentiable module that links spike firing to discrete aggregation.

In SDA, a LIF neuron is assigned to each spatial coordinate and polarity combination (x, y, p), ensuring typical aggregation behaviors. Given a sampled event set $\hat{E}$, an incoming event is selected for aggregation only if it drives its corresponding neuron to fire a spike. Aggregation is then achieved by accumulating the membrane potential $\boldsymbol{u}$ of each event in the selected event set.

Since spikes are triggered by the dynamic evolution of membrane potential, selection and aggregation in SDA are naturally coupled into a unified operation:

$$\hat{E}(x,y,p) = \{e_i | (x_i, y_i, p_i) = (x, y, p) \wedge e_i \in \hat{E}\} \quad (7)$$

$$g_{SDA}(x,y,p) = \sum_{\substack{(x_i,y_i,p_i,t_i)\in\hat{E}(x,y,p) \\ \wedge\, \boldsymbol{s}_{t_i}(x_i,y_i,p_i)=1}} \boldsymbol{u}_{t_i}(x_i,y_i,p_i) \quad (8)$$

where $\boldsymbol{s}_t$ denotes the spike firing state of neuron (x, y, p) at time t, shown in Eq. (5).

An illustration of SDA is shown in Fig. 3. Unlike previous SNN-based aggregation methods (referred to as Spiking Continuous Aggregation, SCA) [36], SDA explicitly distinguishes event importance and selectively incorporates only informative events into aggregation, leading to stronger adaptability and enhanced filtering capability.

By applying surrogate gradients in backpropagation through time (BPTT), SDA can be jointly trained with downstream networks. It should be noted that due to current hardware constraints, μs-level event data is approximated using T_a slices obtained via early Event Count, as shown in Eq. (3), with a fixed temporal resolution Δt_a, *i.e.*, the sampled event set $\hat{E}$ has the shape $T_a \times 2 \times H \times W$.

4.2. Gated Recurrent Spiking Neuron

Effective spiking discrete aggregation in event-based detection requires the ability to assign appropriate importance to events across time, especially under noisy or sparse conditions. However, LIF neurons rely solely on simple integrate-and-fire dynamics, lacking the capacity to adaptively capture complex temporal dependencies. To address this limitation, we enhance LIF neurons by incorporating recurrent connections and gating mechanisms.

We replace the constant decay factor τ in the LIF neuron (as shown in Eq. (4)) with a learnable reset gate $\boldsymbol{R}$, allowing the neuron to adaptively control the influence of past membrane potential states on the present update:

$$\boldsymbol{u}_t(x,y,p) = \boldsymbol{R}_t(x,y,p) \cdot \hat{\boldsymbol{u}}_{t-1}(x,y,p) + \boldsymbol{I}_t(x,y,p) \quad (9)$$

where $\cdot$ denotes element-wise multiplication. When $\boldsymbol{R}_t$ close to 0, prior state information is discarded, effectively filtering out accumulated noise.

In addition, we also introduce gating and recurrent connections on the input current:

$$\begin{aligned} \boldsymbol{I}_t(x,y,p) &= \boldsymbol{F}_t(x,y,p) \cdot \boldsymbol{I}_{t-1}(x,y,p) + \boldsymbol{c}_t(x,y,p) \\ \boldsymbol{c}_t(x,y,p) &= W_I^q \boldsymbol{q}_t(x,y,p) + W_I^s \boldsymbol{s}_{t-1}(x,y,p) \end{aligned} \quad (10)$$

where $\boldsymbol{s}_{t-1}$ is the spike output at the last time step, and $\boldsymbol{F}$ denotes the forget gate that modulates how much of the previous input current should be retained. W_I^q and W_I^s are learnable weights applied to the input $\boldsymbol{q}_t$ and the previous spike $\boldsymbol{s}_{t-1}$, respectively.

The reset gate $\boldsymbol{R}$ and forget gate $\boldsymbol{F}$ are computed as follows, allowing the neuron to modulate both internal mem-

ory and external input in a dynamic, learnable manner:

$$\begin{aligned}\boldsymbol{R}_t(x,y,p) &= \sigma(W_R^q \boldsymbol{q}_t(x,y,p) + W_R^s \boldsymbol{s}_{t-1}(x,y,p)) \\ \boldsymbol{F}_t(x,y,p) &= \sigma(W_F^q \boldsymbol{q}_t(x,y,p) + W_F^s \boldsymbol{s}_{t-1}(x,y,p))\end{aligned} \tag{11}$$

where σ is the sigmoid function.

Furthermore, considering the strong spatial correlation of events within local regions, a 3×3 convolutional kernel is employed for all the weight computations mentioned above, *e.g.*, $W_F^q \boldsymbol{q}_t(x,y,p) = Conv_F^q(\boldsymbol{q}_t(x,y,p))$.

4.3. SDA with Multi-Timescale Fusion

Given the importance of multi-timescale features in event-based object detection, we propose the Multi-Timescale Fusion (MTF) method, which enhances SDA by incorporating coarse-grained temporal information.

The coarse-grained temporal features (additional timescales) are supported by SCA with adaptive sampling (SAS). For detailed methodology, please refer to Wang *et al*. [36]. In brief, SAS samples and continuously aggregates events between two adjacent spike firing times, $t_{j-1}(x,y,p)$ and $t_j(x,y,p)$. The sampled event set is formulated as follows:

$$\hat{E}_j^A(x,y,p) = \left\{ e_i \,\middle|\, \begin{array}{l} t_i \in \left[t_{j-1}(x,y,p), t_j(x,y,p)\right] \\ \wedge\, (x_i,y_i,p_i) = (x,y,p) \end{array} \right\} \tag{12}$$

Within each sampled interval, both SDA and SAS perform their respective operations, and the output of SAS is computed as follows:

$$f_{SAS}(x,y,p) = \sum_{e_i \in \hat{E}_j^A(x,y,p)} \boldsymbol{u}_{t_i}(x_i,y_i,p_i) \tag{13}$$

The aggregated representations from SDA and SAS at different timescales are fused by element-wise addition and then fed into the downstream detection network. However, we observe that simply fusing the two representations by addition does not consistently improve detection performance (Tab. 3, rows 5-6). To address this, we leverage the membrane potentials from the SAS branch to generate a soft mask $\boldsymbol{M}$ within each sampled interval and incorporate it into the current update of gated recurrent spiking neurons in SDA. The computation is defined as follows:

$$\boldsymbol{M}_j(x,y,p) = \sigma\Big(\sum_{e_i \in \hat{E}_j^A(x,y,p)} \boldsymbol{u}_{t_i}(x_i,y_i,p_i)\Big) \tag{14}$$

$$\begin{aligned}\boldsymbol{I}_t(x,y,p) = (\boldsymbol{F}_t \cdot \boldsymbol{I}_{t-1})(x,y,p) + \boldsymbol{M}_j(x,y,p) \cdot \boldsymbol{c}_t(x,y,p) \\ \textbf{\textit{where}}\ t_{j-1}(x,y,p) \le t \cdot \Delta t_a < t_j(x,y,p)\end{aligned} \tag{15}$$

The overall process is illustrated in Fig. 2B. In practice, the slice interval Δt_a used in SDA and the segment interval Δt_m used in SAS ($\Delta t_a \le \Delta t_m$) represent different timescales. In addition, the varying sampled intervals produced by adaptive sampling further provide SDA with richer temporal features.

5. Experiments

5.1. Experimental Setup

Datasets. We evaluate our model on three event-based datasets: Gen1 [9], 1Mpx [29] and N-Caltech101 [30]. The Gen1 dataset is a widely used benchmark for event-based object detection, recorded with a 304×240 resolution event camera. The 1Mpx dataset comprises 14.65 hours of high-resolution event recordings at 1280×720 pixels. The N-Caltech101 is a neuromorphic version of the Caltech101, comprising 8,709 event streams across 101 object categories, each annotated with a single bounding box.

Implementation Details. We train all models using the Adam optimizer with an initial learning rate of 0.002 and a cosine decay schedule. Dataset preprocessing and augmentation follow prior works RVT [11] and EAS-SNN [36]. For Gen1, we use the event stream within 240 ms before each annotation; for 1Mpx, we adopt the 3 segments provided by RVT. Our detection framework uses fully spiking (We define "fully spiking" as the setting where all neurons in the pipeline are spiking neurons) implementations of YOLOX-M (25.3M) and YOLOX-S (8.9M), where the Backbone, FPN, and Head are built using P-LIF and SEW-Residual blocks with 3 time steps, consistent with EAS-SNN. All models are evaluated using COCO $\text{mAP}_{50:95}$ and mAP_{50}.

5.2. Performance Comparison

We evaluate our method on Gen1, 1Mpx, and N-Caltech101. For clarity, we use DASNN to denote the model with only SDA and DASNN-MTF to denote the model with SDA-MTF.

According to Tab. 1, on the Gen1 dataset, DASNN-MTF achieves the best performance of 43.4% $\text{mAP}_{50:95}$, surpassing the state-of-the-art fully spiking model (SpikeYOLO) by 4.5%. Even under a smaller model size (8.9M parameters), it still outperforms the SpikeYOLO by 1.6%. In particular, compared to EAS-SNN, which shares an identical detection architecture, DASNN achieves a 5.3% gain with only half parameters in event representation, which proves the effectiveness of the proposed SDA method. Notably, although the inherently sparse and spike-driven dynamics of SNNs pose challenges for regression tasks, DASNN-MTF still achieves performance exceeding 43% $\text{mAP}_{50:95}$, matching the performance level of ANNs.

To the best of our knowledge, this is the first work to train and evaluate a fully spiking model on 1Mpx. Accord-

Table 1. Performance comparison on the Gen1 dataset. The letters M and S in the parentheses (·) indicate medium and small model scales based on YOLOX-M/S. The slice interval Δt_a for SDA is 20 ms and the segment interval Δt_m for SAS is 60 ms.

Method	Representation	Backbone	Temporal	T	Params	$mAP_{50:95}$	mAP_{50}
AEGNN [32]	Graph	GNN	No	-	20.0M	16.3	-
ASTMNet [18]	1D TCNN	CNN+RNN	Yes	3	>100M	46.7	-
RVT [11]	HIST.	Transformer+RNN	Yes	21	18.5M	47.2	-
Events+YOLOv3 [16]	HIST.	CNN	No	-	>60M	31.2	-
SMamba [39]	HIST.	SSM+RNN	Yes	21	16.1M	**50.4**	-
LT-SNN [15]	HIST.	SNN	Yes	-	-	29.8	-
EMS-ResNet34 [34]	HIST.	SNN	Yes	5	14.4M	31.0	59.0
SFOD [34]	Voxel Cube	SNN	Yes	-	11.9M	32.1	59.3
EAS-SNN†[36]	ARSNN	SNN	Yes	3	25.3M	37.5	69.9
SpikeYOLO [23]	HIST.	SNN	Yes	5	23.1M	38.9	67.2
KD-SNN [2]	HIST.	SNN	Yes	5	13.0M	22.9	-
CREST [24]	MESTOR	SNN	Yes	5	7.61M	36.0	63.2
Ours(S)	**SDA**	SNN	Yes	3	8.9M	39.9	69.9
	SDA-MTF		Yes	3	8.9M	40.5	70.5
Ours(M)	**SDA**	SNN	Yes	3	25.3M	42.8	72.6
	SDA-MTF		Yes	3	25.3M	**43.4**	**73.1**

† denotes the full spiking version of EAS-SNN.

Table 2. Performance comparisons on 1Mpx and N-Caltech101.

Method	Net	1Mpx		N-Caltech101	
		$mAP_{50:95}$	mAP_{50}	$mAP_{50:95}$	mAP_{50}
RED	ANN	43.0	-	-	-
RVT	ANN	47.4	-	-	-
Events+YOLOv3	ANN	34.6	-	-	-
AEGNN	ANN	-	-	-	59.5
EAS-SNN†	SNN	28.1	55.3	36.3	56.4
DASNN-MTF(M)	SNN	30.4	59.1	40.1	61.9

† denotes the full spiking version of EAS-SNN.

ing to Tab. 2, DASNN-MTF achieves over 30% $mAP_{50:95}$, demonstrating the effectiveness of both our method and SNNs on a large-scale event-based dataset. On the N-Caltech101 dataset, DASNN-MTF outperforms the state-of-the-art fully spiking method EAS-SNN by 5.5% mAP_{50}.

5.3. Ablation Studies

To evaluate the effectiveness of our proposed method, we conduct a series of experiments on the Gen1 dataset with spiking YOLOX-M.

Spiking Discrete Aggregation. Since SDA performs finer-grained aggregation for events, it operates with a higher temporal resolution compared to continuous aggregation in practice. To rule out the possibility that the observed performance improvement is merely due to higher temporal resolution, we adopt the same model architecture trained with Spiking Continuous Aggregation (SCA) as our baseline and set the same temporal resolution (20 ms). The results shown in Tab. 3 suggest that SDA achieves better performance, with an improvement of 1.6% $mAP_{50:95}$. This indicates that by aggregating informative events, SDA provides more discriminative spatiotemporal representations for detection.

Table 3. Comparison with common event representations and ablation of SDA and SDA-MTF on the Gen1 dataset.

Method	GRSN	Mask	$mAP_{50:95}$	mAP_{50}
Event Count	–	–	32.3	58.0
Time Surface	–	–	37.3	66.5
Voxel Cube	–	–	32.6	59.2
Baseline *	✓	–	41.2	71.3
SDA *	✓	–	42.8	72.6
SDA-MTF ‡	✓	✗	42.8	72.8
SDA-MTF	✗	✗	40.7	70.1
SDA-MTF	✓	✓	43.4	73.1

* Temporal resolution is 20 ms.
‡ All SDA-MTF variants use $\Delta t_m = 60$ms and $\Delta t_a = 20$ms.

Gated Recurrent Spiking Neuron and Soft Mask. In this ablation study, we analyze the effect of Gated Recurrent Spiking Neuron (GRSN) and the role of soft mask in MTF. According to Tab. 3, simply adding representations from

Table 4. Compatibility of SDA and SDA-MTF when integrated into various ANN-based detectors on the Gen1 dataset. The results demonstrate that incorporating SDA and SDA-MTF consistently improves performance across different architectures, including CNN, Transformer and Transformer+RNN.

Method	Backbone	Representation	$mAP_{50:95}$
YOLOX-S	CNN	Event Count	37.4
		+ SDA	45.8(**+ 8.4**)
		+ SDA-MTF	43.3(**+ 5.9**)
PVT-S [35]	Transformer	Event Count	42.4
		+ SDA	45.4(**+ 3.0**)
		+ SDA-MTF	45.2(**+ 2.8**)
RVT-B	Transformer+RNN	Event Count	47.2
		+ SDA	48.4(**+ 1.2**)
		+ SDA-MTF	48.7(**+ 1.5**)

different temporal scales fails to improve detection performance. However, when the mask is used to regularize the SDA with GRSN, a performance gain is observed. In addition, without GRSN, the performance of SDA-MTF drops by more than 2%.

Comparison with Common Event Representations. We compare SDA and SDA-MTF with three widely adopted event representations: Event Count, Time Surface and Voxel Cube. Tab. 3 confirms the significant advantage of the proposed spiking discrete aggregation over these representation methods.

5.4. Experimental Analysis

Compatibility with ANNs. While the previous comparisons demonstrate that SDA and SDA-MTF achieve high performance within SNNs, we further explore their compatibility with ANNs. To this end, we adopt three representative detection architectures (YOLOX-S, PVT-S, and RVT-B) as ANN baselines to evaluate the effectiveness of SDA and SDA-MTF. The input event stream is pre-aggregated based on Event Count. For YOLOX-S and PVT-S, we use a 20 ms slice interval and a 240 ms temporal window, while for RVT-B, we follow the original settings of RVT [11].

As shown in Tab. 4, integrating SDA and SDA-MTF into YOLOX-S leads to significant improvements of 8.4% and 5.9% in $mAP_{50:95}$, respectively. For PVT-S, incorporating SDA and SDA-MTF yields gains of +3.0% and +2.8%, demonstrating that our approach also benefits Transformer-based architectures. When applied to RVT-B, which already models temporal dependencies through recurrent connections, the improvements are relatively smaller (+1.2% and +1.5%), yet still indicate the compatibility of SDA and SDA-MTF with temporal backbones.

These results demonstrate that SDA and SDA-MTF not only benefit SNNs but also generalize effectively to conventional ANN-based detectors.

Table 5. Evaluation across speed levels on Gen1 ($mAP_{50:95}$). A higher level corresponds to a faster object motion speed.

Method	Lv1	Lv2	Lv3	Lv4	Overall
SDA	23.9	47.8	47.9	42.5	42.8
SDA-MTF	**24.3**	**48.1**	**48.8**	**44.0**	43.4

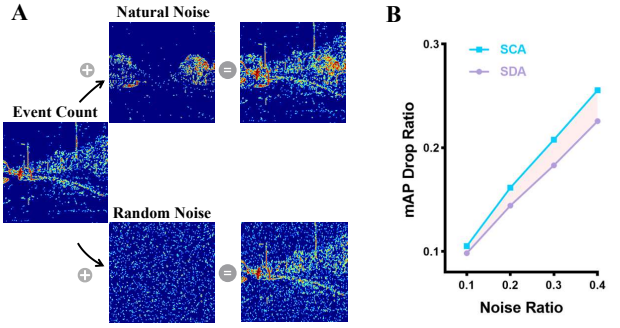


Figure 4. **A.** Construction of the noise test dataset. Natural and random noise are added to the original event data on Gen1 to simulate different noisy conditions. **B.** Comparison of performance drop ratios between SCA and SDA under varying noise ratios.

Evaluation Under Different Motion Speeds. According to Liu *et al*. [20], we classify objects in Gen1 into four speed levels by computing optical flow. As shown in Tab. 5, SDA-MTF yields performance gains across all levels. The benefits of SDA-MTF are evident in high-speed scenarios, as reflected by larger gains on Lv3 and Lv4.

Impact of Temporal Resolution. We evaluate DASNN-MTF(M) on Gen1 with different slice intervals Δt_a. Results are provided in the supplementary material.

Robustness to Noisy Events. To validate the robustness of SDA under noisy scenarios, we construct a noise test dataset comprising both natural and random noise, as shown in Fig. 4A. Implementation details are provided in the supplementary material. We evaluate the performance degradation of SCA and SDA under four noise levels, defined by the ratio of noisy events to original events. As shown in Fig. 4B, SDA consistently shows lower $mAP_{50:95}$ drop ratio across all noise levels. Notably, as the noise ratio increases from 0.1 to 0.3, the performance gap between the two methods widens, demonstrating the superior filtering capability of our method.

Qualitative Analysis. We perform visualizations on the Gen1 dataset and select scenarios involving cars and pedestrians with varying degrees of motion blur. As presented in Fig. 5, the top row displays raw Event Count without SDA, where object boundaries are obscured by noisy

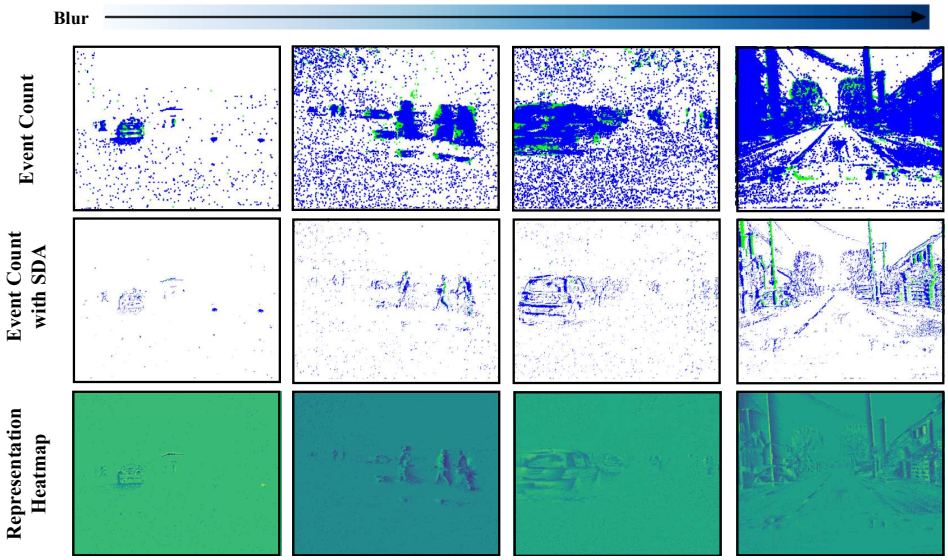


Figure 5. Visualizations of Event Count (without and with SDA) and the SDA-MTF representation heatmap under increasing blur.

Table 6. Comparison of energy consumption on the Gen1 dataset.

Module	YOLOX-S		EAS-SNN(S)			**DASNN(S)**			**DASNN-MTF(S)**		
	MAC(G)	Energy(mJ)	AC(G)	MAC(G)	Energy(mJ)	AC(G)	MAC(G)	Energy(mJ)	AC(G)	MAC(G)	Energy(mJ)
Embedding	0	0	0.02	1.63	7.52	0.70	1.19	6.10	0.10	1.32	6.16
Backbone	5.24	24.10	2.66	0.71	5.64	3.37	0.71	6.30	3.36	0.71	6.29
FPN	2.63	12.09	1.26	0	1.14	1.45	0	1.31	1.54	0	1.39
Head	5.04	23.17	1.79	0	1.61	2.17	0	1.95	2.48	0	2.23
Total	12.90	59.35 (1×)	5.74	2.34	15.91 (3.73×)	7.69	1.90	**15.66 (3.79×)**	7.49	2.03	16.08 (3.69×)

events. The middle row shows the Event Count after SDA. As uninformative events are filtered out, object contours become clearer, demonstrating SDA's ability to aggregate informative events. The bottom row presents the corresponding representation heatmaps from SDA-MTF, which preserves object shapes despite heavy motion blur.

5.5. Energy Efficiency Analysis

SNNs utilize binary spikes, leading to energy-efficient accumulation (AC) operations on neuromorphic hardware, unlike the dense multiplication-and-accumulation (MAC) operations used by ANNs. Notably, since the recurrent connections inside spiking neurons and the soft mask in MTF introduce extra computation, we estimate their energy using the same cost as a MAC operation. The detailed calculations are shown in the supplementary material.

Energy consumption is evaluated on the Gen1 dataset with SNNs configured to run for $T = 3$ timesteps. As shown in Tab. 6, our proposed DASNN(S) and DASNN-MTF(S) achieve 3.79× and 3.69× reduction in energy consumption compared to the ANN-based YOLOX-S. Furthermore, even compared to EAS-SNN(S), which shares a similar architecture, our DASNN(S) achieves lower energy consumption (15.66 mJ vs. 15.91 mJ).

6. Conclusion

This work proposes a novel event aggregation method, Spiking Discrete Aggregation (SDA), for event-based object detection. Built by gated recurrent spiking neurons, SDA exploits the spike-firing mechanism of SNNs to adaptively aggregate informative events. Enhanced by coarse-grained temporal context via Multi-Timescale Fusion, SDA produces robust representation to improve detection accuracy. Experiments on the three neuromorphic datasets show that SDA and SDA-MTF outperform existing fully spiking models while using fewer parameters. Ablation studies further confirm that the performance gains stem from the SDA mechanism itself. Moreover, we demonstrate their compatibility with ANNs and robustness under noisy conditions.

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